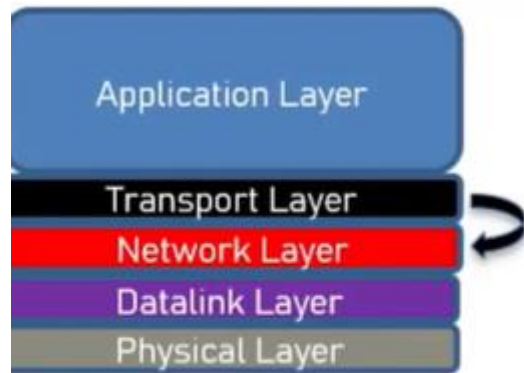


IP-Security (ESP)



Encapsulation Security Protocol (ESP)

ESP can also work in either Transport or Tunnel mode



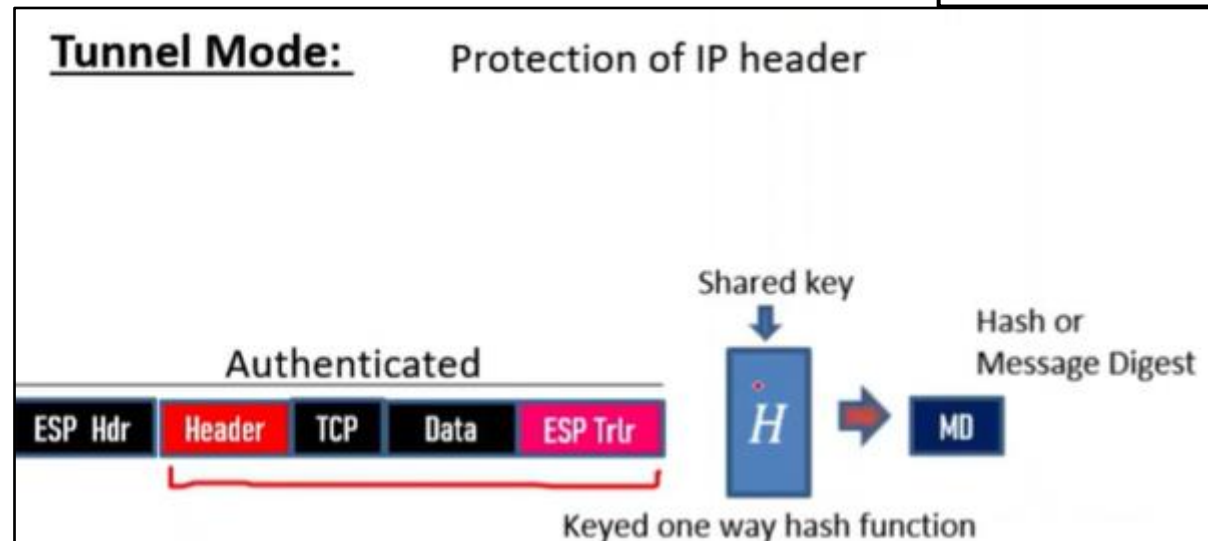
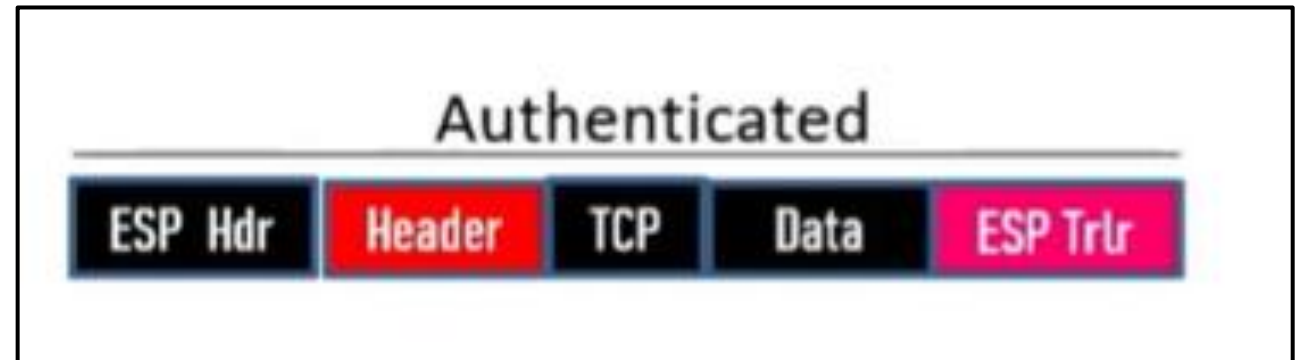
- Header contains the source addr and the destination addr.



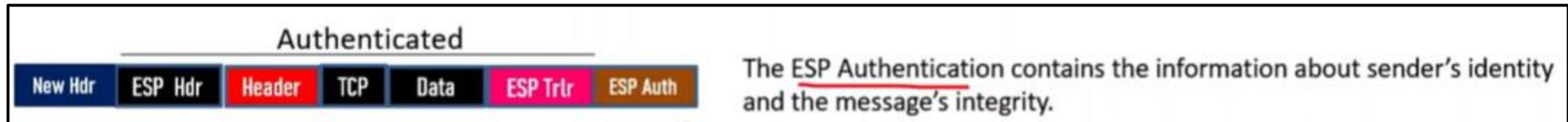
Encryption can be performed using AES, DES, 3DES

ESP Trailer has fields like padding, padding length and next header

- The encrypted message is added with the **ESP Hdr** and then the authentication is performed.

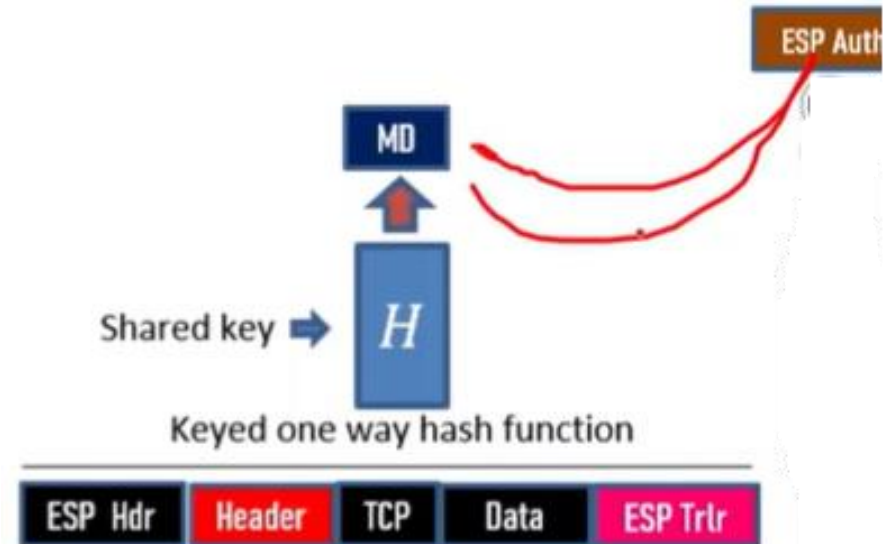


- Once the message digest (MD) is calculated, New Hdr is added, and the MD is added in the ESP Auth field.



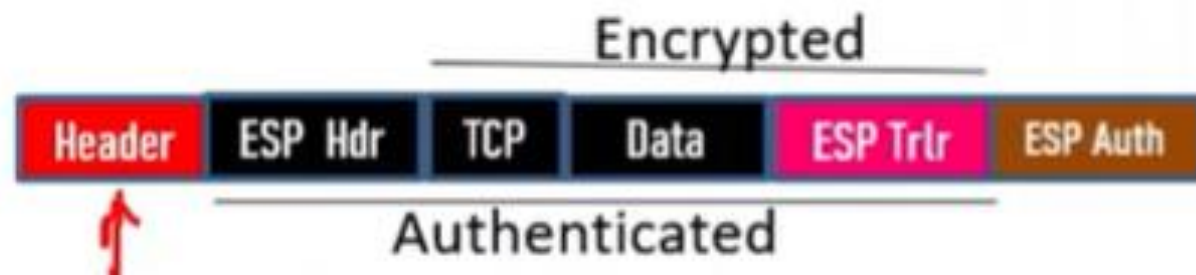
- In the destination side it will generate the message digest and compares the MD with ESP Auth which has the MD generated by the sender. If it matches, then the data will be accepted otherwise the data will be rejected.

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Encapsulation Security Protocol (ESP)

Transport Mode: No protection of IP header



Authentication Header (AH)

It uses IP protocol 51

Encapsulation Security Protocol (ESP)

It uses IP protocol 50

Version	Length	DS Field	Packet Length	
Identification			Flags	Fragment Offset
Time to Live	Protocol		Header Checksum	
Source IP Address				
Destination IP Address				
Options				Padding